Session 7: UML - Sequence and Activity Diagram Modeling

Project: Digital Examination Management System (DEMS)

Deliverable: Sequence Diagram and Activity Diagram

1. Objective

This lab models the behaviour of the Digital Examination Management System using two UML behavioural diagrams. The sequence diagram shows how the system's components exchange messages while a student attempts an online exam. The activity diagram shows the end-to-end exam workflow, including its decisions and the roles involved.

Learning outcomes covered in this session:

- Design a sequence diagram with lifelines, synchronous messages and return messages.
- Represent message flow between the user interface, server, database and evaluation component.
- Model a workflow using an activity diagram with start and end nodes, actions, decisions and merges.
- Relate both diagrams to how they are used in industry.

2. System context

The Digital Examination Management System digitises the full exam lifecycle: student registration and login, question bank management, exam scheduling, timed online exams, automatic grading of objective questions, teacher grading of subjective answers, result generation and report cards. The three user roles are Admin, Teacher and Student. The diagrams in this document focus on the most important interaction: a student logging in, taking an exam and receiving a result.

Participants used in the diagrams

Participant	Type	Responsibility
Student	Actor	Logs in, starts the exam, answers questions and submits.
Exam portal	User interface	Web front end that takes input and displays the paper, timer and result.
Exam server	Application logic	Authenticates users, serves question papers, receives answers and coordinates evaluation.
Database	Data store	Stores users, questions, responses and marks.
Evaluator	Component	Auto-grades objective answers and returns the score.
Teacher	Actor	Grades subjective answers (shown in the activity diagram).

3. Sequence diagram: student attempts an exam

A sequence diagram shows objects as vertical lifelines and the messages between them in time order, from top to bottom. Solid arrows are requests (synchronous calls) and dashed arrows are return messages.

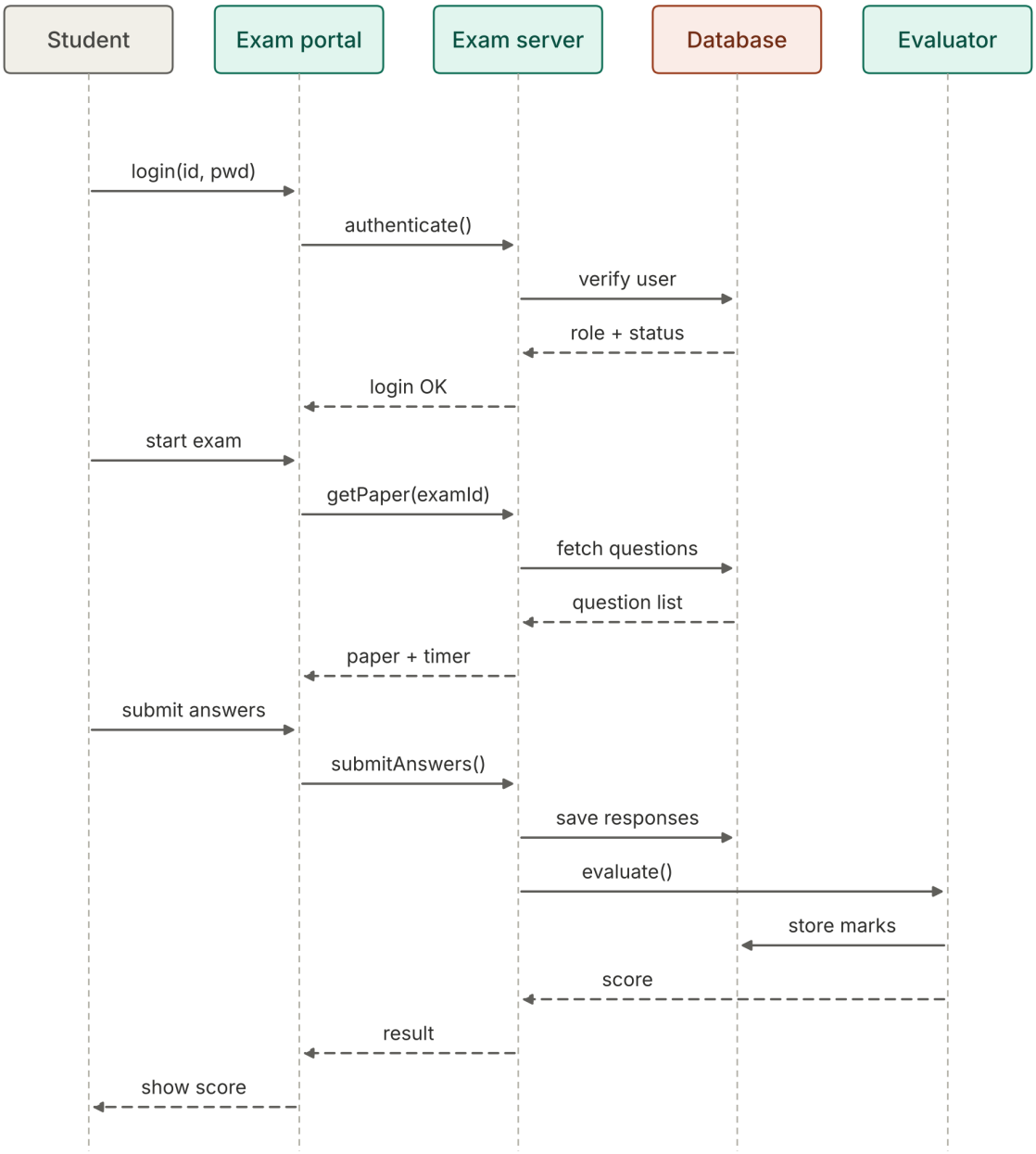


Figure 1: Sequence diagram for the online exam attempt

3.1 Message flow

The numbered messages below match the arrows in Figure 1, read from top to bottom.

#	From	To	Message	Purpose
1	Student	Exam portal	login(id, pwd)	Student enters credentials.
2	Exam portal	Exam server	authenticate()	Portal forwards the credentials.
3	Exam server	Database	verify user	Server looks up the user record.
4	Database	Exam server	role + status (return)	User role and account status come back.
5	Exam server	Exam portal	login OK (return)	Student is signed in.
6	Student	Exam portal	start exam	Student picks the scheduled exam.
7	Exam portal	Exam server	getPaper(examId)	Portal requests the question paper.
8	Exam server	Database	fetch questions	Server reads the questions for the exam.
9	Database	Exam server	question list (return)	Questions are returned.
10	Exam server	Exam portal	paper + timer (return)	Paper is shown and the countdown starts.
11	Student	Exam portal	submit answers	Student submits, or the timer auto-submits.
12	Exam portal	Exam server	submitAnswers()	Answers are sent to the server.
13	Exam server	Database	save responses	Responses are stored.
14	Exam server	Evaluator	evaluate()	Server asks for grading.
15	Evaluator	Database	store marks	Marks are saved.
16	Evaluator	Exam server	score (return)	Score is returned.
17	Exam server	Exam portal	result (return)	Result is passed to the portal.
18	Exam portal	Student	show score (return)	Student sees the score.

3.2 Explanation

- **Authentication (messages 1-5):** the portal never talks to the database directly. The server checks the credentials and returns the role, which later decides what the user may access.
- **Loading the exam (6-10):** the server fetches the questions and returns them with a timer, so the time limit is enforced by the system and not left to the student.
- **Submission and grading (11-16):** answers are saved first and then evaluated. The evaluator stores marks in the database and returns the score to the server.
- **Result (17-18):** the score flows back through the portal to the student.

4. Activity diagram: online exam workflow

An activity diagram models a workflow as a flow of actions. The filled circle is the start node, the ringed circle is the end node, rounded boxes are actions and diamonds are decisions. Colours show who performs each step (student, system or teacher).

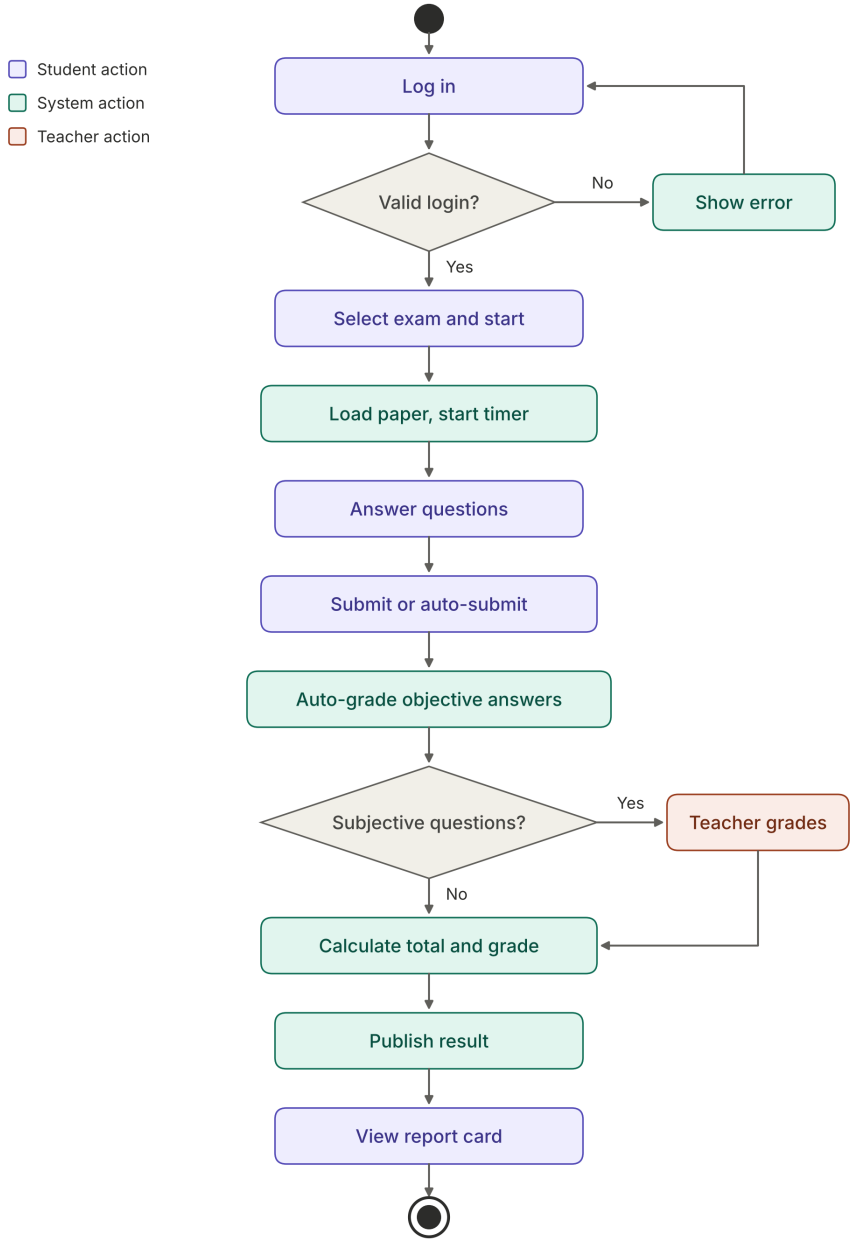


Figure 2: Activity diagram for the exam workflow

4.1 Workflow description

Step	Performed by	Description
Log in	Student	Student enters credentials.
Valid login?	Decision	If invalid, the system shows an error and the student retries. If valid, the flow continues.
Select exam and start	Student	Student opens a scheduled exam.
Load paper, start timer	System	Question paper is loaded and the countdown begins.
Answer questions	Student	Student attempts the questions within the time limit.
Submit or auto-submit	Student / System	Student submits, or the system submits automatically when time is up.
Auto-grade objective answers	System	MCQ and similar answers are marked automatically.
Subjective questions?	Decision	If yes, a teacher grades them. If no, the flow goes straight to result calculation.
Teacher grades	Teacher	Teacher awards marks for descriptive answers.
Calculate total and grade	System	Marks are combined into a total, percentage and grade.
Publish result	System	Result is made visible to the student.
View report card	Student	Student views the result and report card.

4.2 Key modelling points

- **Decision nodes** make the workflow conditional: login can fail and loop back, and the teacher step only happens when the exam has subjective questions.
- **Merge:** both paths (with and without teacher grading) join again at "Calculate total and grade".
- **Roles:** separating student, system and teacher actions shows who owns each step, which is the same idea as swimlanes.

5. Industry mapping: interaction and workflow modeling

The two diagrams answer different questions, and industry teams use them at different stages of building a system.

	Sequence diagram	Activity diagram
Models	Interaction: who sends which message to whom, and in what order.	Workflow: the steps, decisions and parallel paths of a process.
Typical use in industry	Designing APIs and service calls, documenting microservice interactions, debugging integration issues, explaining login and payment flows to developers.	Business process modeling, requirement analysis with stakeholders, designing test cases, planning automation and approval workflows.
Example in this project	Student, portal, server, database and evaluator exchanging messages during an exam.	Login, exam, grading, publishing and report card workflow.
Main audience	Developers and architects.	Analysts, testers and business stakeholders.

Real-world examples: online assessment platforms and university exam portals follow the same pattern of authentication, timed paper delivery, auto-evaluation and result publishing. Teams typically draw the activity diagram first to agree on the process, then use sequence diagrams to design the exact system calls behind each step.

6. Conclusion

The sequence diagram made the message flow between the portal, server, database and evaluator explicit, and the activity diagram captured the full exam workflow with its decisions and roles. Together they give a clear behavioural model of the Digital Examination Management System that can guide implementation and testing.